

FINAL EXPLANATION: BIOLOGY GRADE 11

Teacher Key: Exemplar Answers and Marking Guide

FINAL EXPLANATION: BIOLOGY GRADE 11 – SUB-STRAND 1.4 CELL DIVISION – FINAL EXPLANATION TO THE PHENOMENON

Teacher Key: Exemplar Answers (do not give to students)

INSTRUCTIONS GIVEN TO STUDENTS

Key question: How does the body produce the enormous number of new cells needed for growth? Think back to the baby photographs and the 14-year growth time-lapse from Lesson 1, where a newborn of about 3 kg became a teenager of 50 kg or more. Use what you observed and modelled in all ten lessons to write your Final Explanation. Answer each prompt in the space below it. Write in full sentences, use correct scientific terms, and support each claim with evidence from our investigations (diagrams, clay models, onion root tip squashes, animations, case studies). Compare your final answer with your first model from Lesson 1 at the end.

Part 1: Claim - Where do the new cells come from?

A newborn of about 3 kg becomes a teenager of 50 kg or more, yet we only eat food such as ugali, sukuma wiki and beans. Write a claim that answers the key question: where do the billions of new cells come from? In your answer, say what food does and does not do in growth. Then say how your thinking has changed since your Lesson 1 drawing.

Claim: The body makes the billions of new cells needed for growth through cell division. Existing cells divide to form new cells, starting from the first cell (the zygote). Food such as ugali, beans and sukuma wiki does not turn directly into cells. It supplies the energy and raw materials that cells need to grow and divide. In Lesson 1 I thought cells might just grow bigger or be made from food. Now I know that the increase in the number of cells comes from division of existing cells.

Part 2: Evidence from mitosis - Growth and the right instructions

Use evidence from our diagram analysis, the clay model of 1 cell to 2 cells, and the onion (kitunguu) root tip squashes. (a) Describe what happens to the chromosomes before and during mitosis, naming the stages. (b) Explain how mitosis gives each new cell the right

(a) Before mitosis, in interphase, the DNA is copied, so each chromosome has two identical chromatids joined at a centromere. In prophase the chromosomes become visible, in metaphase they line up in the middle of the cell, in anaphase the chromatids separate and move to opposite ends, and in telophase two new nuclei form. Cytokinesis then divides the cell into two. (b) Because the chromosomes are copied and shared accurately, each daughter cell receives a full set of chromosomes with the same genetic information (DNA and genes) as the parent cell, so it has the right instructions to work properly. (c) Growing tissue such as a root tip has cells in the process of dividing, so new cells are being made there. At any moment most cells are in interphase and a smaller number are dividing, but dividing cells are present, which is evidence that growth happens by division. (d) Each round of mitosis doubles the number of cells (1, 2, 4, 8, 16...). After many rounds, starting from one zygote, this doubling reaches billions of cells.

instructions. (c) In the root tip squashes, most cells were in interphase and fewer were dividing. What does this tell you about growing tissue? (d) Explain how repeated division (1, 2, 4, 8, 16...) can lead to billions of cells.	
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Part 3: Evidence from meiosis and other forms of cell division	
(a) A sperm and an egg each come from a body cell with 46 chromosomes. Explain, using the calculation we did in Lesson 4, why meiosis is needed and how many chromosomes each gamete has. (b) Describe what meiosis produces compared with mitosis. (c) Name two ways a single organism can make new organisms or cells from one parent without sex cells, giving a living example of each, and say how they are similar.	(a) Without meiosis, a sperm with 46 chromosomes joining an egg with 46 chromosomes would make a zygote with $46 + 46 = 92$ chromosomes, and the number would double every generation. Meiosis halves the number, so each gamete has 23 chromosomes. When they fuse at fertilisation the zygote has 46 chromosomes again and can then divide by mitosis to grow into a child. (b) Mitosis produces two genetically identical cells with the full chromosome number and is used for growth and repair. Meiosis happens in the reproductive organs, has two divisions (meiosis I and II), and produces four cells with half the chromosome number and different combinations of genes, used for sexual reproduction. (c) Budding, as in yeast (for example in mandazi dough or busaa), where a small bud grows and separates from the parent. Binary fission, as in bacteria such as Salmonella or E. coli, where the cell copies its chromosome, elongates and splits into two equal cells. Both are asexual reproduction from one parent with no fusion of sex cells, both copy the genetic material before dividing, and both increase numbers quickly.

Part 4: Why cell division matters	
Answer the KICD inquiry question: How is cell division important? Use at least three examples from our case studies, including the healing cut, the maize or sukuma wiki plant, and tissue culture of bananas or tea in Kenya. For each example, state whether mitosis or meiosis is involved and why.	Cell division is important for growth, repair, reproduction and development. Growth: a child grows because body cells divide by mitosis to make more cells. Repair: when a skin cut heals, cells near the wound divide by mitosis to replace the damaged ones. Plants: a maize or sukuma wiki plant grows new roots and shoots by mitosis at growing tips, and some plants can reproduce asexually. Tissue culture: in Kenya, farmers and KALRO use tissue culture to multiply bananas and tea. A small piece of plant (explant) grows into a callus, then shoots and roots, by repeated mitosis in a sterile nutrient medium, making many identical disease-free plantlets quickly. Reproduction: meiosis makes the gametes for sexual reproduction, which keeps the chromosome number constant and gives offspring that differ from their parents. So cell division is essential for the growth, development, repair and reproduction of living organisms.

Part 5: Final Explanation - Claim, Evidence, Reasoning

Write your complete Final Explanation to answer the key question: How does the body produce the enormous number of new cells needed for growth? Use the Claim-Evidence-Reasoning structure. Include (1) your claim, (2) at least three pieces of evidence from our investigations (including mitosis and meiosis), and (3) reasoning that links the evidence to the growth of a child from 3 kg to 50 kg or more and to the wider importance of cell division.	Claim: The body produces new cells through cell division. Evidence: In our diagrams and clay model, one parent cell divided to form new cells, and the counts in our table showed the number of cells increasing. In our mitosis diagram and onion root tip squashes we saw chromosomes become visible, line up, separate and move into two new nuclei, with some cells dividing in the growing tip. Each daughter cell received the same chromosomes as the parent. In meiosis we saw one cell make four cells with half the chromosome number (23), so gametes can fuse to restore 46 chromosomes in the zygote. We also saw yeast budding and bacteria dividing by binary fission, and tissue culture making many plantlets. Reasoning: A child grows from 3 kg to 50 kg or more because the zygote and later body cells divide by mitosis again and again (1, 2, 4, 8, 16...), producing billions of cells. Because DNA is copied before each division, every new cell gets an identical set of chromosomes and so the right instructions. Food such as ugali, beans and sukuma wiki gives energy and raw materials but the new cells come from existing cells dividing. Meiosis makes the gametes that start a new individual. Therefore, cell division is essential for the growth, development, repair and reproduction of living organisms.
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Criterion	Excellent (4)	Proficient (3)	Developing (1–2)
Claim and use of the phenomenon	Gives a clear claim that new cells come from division of existing cells, explains that food gives energy and raw materials but does not become cells, and links directly to the 3 kg to 50 kg growth of a child and to the Lesson 1 model.	States that cell division makes new cells and links to the growth of a child, but the role of food or the link to the Lesson 1 model is brief or incomplete.	Claim is vague or incorrect (for example, food turns directly into cells or cells only grow bigger) and there is little link to the phenomenon.
Understanding of mitosis	Accurately describes DNA copying, the four stages and cytokinesis, explains how each daughter cell gets identical chromosomes and the right instructions, and explains doubling (1, 2, 4, 8, 16...) to billions of cells.	Describes mitosis and the formation of two identical cells with mostly correct stages, but explanation of how instructions are passed on or of doubling is partial.	Describes mitosis with major errors or omissions, such as missing stages or no mention of chromosomes being copied and shared.
Understanding of meiosis and other cell division (budding and binary fission)	Explains with the $46 + 46 = 92$ reasoning why meiosis halves the chromosome number to 23, contrasts it correctly with mitosis, and accurately describes budding and binary fission with examples as asexual reproduction.	Correctly states that meiosis makes gametes with half the chromosomes and names budding and binary fission, but comparison or examples are limited.	Confuses mitosis and meiosis or does not correctly describe gametes, budding or binary fission.
Use of evidence and Kenyan examples	Uses at least three specific pieces of evidence from the investigations (such as the onion root tip squashes, clay model,	Uses two relevant pieces of evidence, with some connection to the claim.	Uses little or no evidence, or gives general statements not taken from our

	case studies, tissue culture of bananas or tea) and ties each to the claim.		investigations.
Reasoning, importance of cell division and scientific communication	Logically links evidence to the claim, explains the importance of cell division for growth, development, repair and reproduction, and uses correct terms in clear, well-organised Claim-Evidence-Reasoning writing.	Reasoning mostly links evidence to the claim and mentions some importance of cell division, with minor errors in terms or organisation.	Reasoning is missing or unclear, few uses of the importance of cell division, and scientific terms are used incorrectly or not at all.

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